

Metals, Nonmetals & Metalloids

Use the cue, then fade the cue. Look at the cue card for 60 seconds. Then cover it before answering Round 1. Check the card only after you commit to an answer.

Cue focus	Metals are usually shiny and conductive, nonmetals are often dull and poor conductors, and metalloids have mixed properties near the periodic table stair-step.
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Round 1 - Retrieve It

1. Name two common properties of metals.

2. How do nonmetals usually compare with metals in electrical conductivity?

3. Where are many metalloids found on the periodic table?

4. Why are metalloids useful in electronics?

Round 2 - Sketch It

Draw a simplified periodic table rectangle. Mark a stair-step line and label the general metal, nonmetal, and metalloid regions.



Metals, Nonmetals & Metalloids - Apply It

Keep the cue card covered. Solve first. Then use the cue to check, revise, and explain what you changed.

Round 3 - Apply the Relationship

1. A sample is shiny, bends without breaking, and conducts electricity well. Best classification?

2. A sample is dull, brittle, and a poor conductor. Best classification?

3. A sample has some metallic shine but only conducts under certain conditions. Best classification?

Round 4 - Reason from Evidence

A device needs a material that can control electrical flow rather than conduct extremely well or block it completely. Which category is the best starting point, and why?

Metals, Nonmetals & Metalloids - ACE It

Use the cue card only if you need it. The goal is to show what you can explain, connect, and use on your own.

A**Articulate**

Explain Metals, Nonmetals & Metalloids in your own words. Use one example that is not printed on the cue card.

C**Connect**

Connect properties to use. Why is copper useful for electrical wiring while many nonmetals are used as insulators?

E**Extend**

An unknown solid is brittle but has moderate conductivity. Predict its category and name one additional test that would strengthen your claim.

Confidence check Circle one after you finish: I still need the cue / I need it once in a while / I can explain this without it

TEACHER KEY - Metals, Nonmetals & Metalloids

Remove this page before student copying. Answers below give expected ideas, not required wording.

Retrieve It

1. Examples: shiny/lustrous, conductive, malleable, ductile.
2. Nonmetals are generally poor conductors.
3. Along the stair-step boundary between metals and nonmetals.
4. They have intermediate or semiconducting electrical properties.

Sketch It - look for

Metals occupy most of the left/center, nonmetals are mainly upper right, and metalloids lie along the stair-step boundary. Exact element placement is not required for this sketch.

Apply It

1. Metal.
2. Nonmetal.
3. Metalloid.

Reason from Evidence - look for

Metalloid, because semiconducting behavior falls between strong conductors and insulators.

ACE - Extend look-for

Likely metalloid. Additional evidence could include location on the periodic table, luster, or measured conductivity compared with known samples.

Suggested use

Run Page 1 with the cue card available, Page 2 with the cue mostly covered, and Page 3 as independent reflection. Let students uncover the cue to self-correct in a different color or with a revision mark.